

INTERPRETABILITY FOR THE SCIENCE OF TRAINING: EMERGENCE, PREDICTION, AND STEERING (STEPS)

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Tagline. Toward actionable training dynamics: describe trajectories, predict outcomes, steer training.

Website. <https://science-of-training.github.io/>

1 WORKSHOP DESCRIPTION

Interpretability research has produced a wide-ranging toolkit, with probing, sparse dictionary learning (Bricken et al., 2023; Bhalla et al., 2026), and data attribution (Koh & Liang, 2020; Liu et al., 2025; Matlin et al., 2026; Kim et al., 2026), but these methods are mostly applied to finished models. How a model gets there is studied through training dynamics and additionally with other prediction methods like scaling laws. Aggregate statistical relationships, such as scaling laws (Kaplan et al., 2020; Hoffmann et al., 2022; Bahri et al., 2024; Caballero et al., 2023; Pearce & Song, 2024) and architecture-based performance estimation (White et al., 2021; Elsken et al., 2019; Mellor et al., 2021), predict final performance, while studies of training dynamics document *when* representations and behaviors emerge, whether over training steps (Chen et al., 2025) or across scale (Wei et al., 2022). These lines of work establish that the path a model takes through training shapes what it becomes. Then, a relevant question, and the focus of this workshop, is *which interpretability methods work for understanding training, and what actions can they enable that outcome-only evaluation cannot?*

A trajectory-level view of training is useful precisely because it surfaces phenomena that final perplexity, accuracy, or win-rates cannot. For example, models may perform well on downstream evaluations while becoming overtrained, making them difficult to fine-tune subsequently (Springer et al., 2025; Dohare et al., 2024). If model flexibility could be quantified *during* training, such failure modes could be anticipated before they manifest downstream. In a similar vein, recent work on the interaction between fine-tuning and pretraining suggests that training dynamics can predict when adaptation will succeed, degrade, or induce forgetting (Ren & Sutherland, 2025; Sun & Dredze, 2025; Bethune et al., 2025). Bridging the gap between understanding and practice requires moving beyond conceptual descriptions toward diagnosing failures that outcome-only evaluation misses and toward guiding interventions during training.

This workshop aims to encourage such types of work that can provide insights toward a dynamical and actionable understanding of training (Biderman et al., 2026; Orgad et al., 2026). We organize this around three escalating goals:

1. **Descriptive:** characterizing how representations, circuits, gradients, and behaviors evolve over the course of training.
2. **Predictive:** identifying signals early in training that reliably forecast later capabilities, failures, generalization, or structural changes.
3. **Prescriptive:** using predictive insight to actively steer, control, and repair training.

[[figl: pair these with interp methods](#)] We organize the workshop’s topics of interest around the three goals detailed below and welcome work that shares similar interests:

1. Descriptive

- Representation formation, phase transitions, reorganization, and interaction across training
- Interpretability across time, scale, gradients, and optimizer dynamics

- Characterizing random variations and initial conditions (van der Wal et al., 2025; Sellam et al., 2022)

2. Predictive

- Early monitoring of training failures and criteria for stopping
- Stress-testing of early checkpoints and identifying misrepresentations that lead to downstream malformation
- Pretraining evaluations and invariants that predict post-training behavior across scale
- Generalization prediction: what transfers from which data, when, and why
- Training outcome prediction, including negative results

3. Prescriptive

- Training repair, such as reverting to and correcting from earlier checkpoints
- Causal and interventional methods for training (Biderman et al., 2023b)
- Phase specific data selection strategies (O’Brien et al., 2025; Tice et al., 2026)
- Controlling model behavior during training

Relevant Workshops Our proposed workshop is closely related to the High-dimensional Learning Dynamics (HiLD) workshop series (HiLD Workshop Organizers, 2023; 2024; 2025; 2026), overlapping with our *descriptive* layer: representation formation, phase transitions, and interpretability across training. However, we substantially broaden this scope to include predicting from early failure signals and possible training repairs —what could make the science of training actionable Orgad et al. (2026). Beyond descriptive analysis, we emphasize *predictive* and *prescriptive* dimensions of training dynamics, seeking work in both theory as well as how the theoretical insights inform practice, beyond HiLD’s theoretical focus. We highlight the need for methods that *anticipate* training dynamics and outcomes (e.g., early detection of failure modes, generalization prediction, and checkpoint stress-testing) and that enable actions such as training repair, causal manipulation, and phase-aware data selection. A central goal is to bring together academic researchers and practitioners to identify which analyses are useful in practice, where current methods fall short, and what predictive or prescriptive tools would best support real-world model development.

2 INVITED SPEAKERS AND PANELISTS

Andrew Saxe <a.saxe@ucl.ac.uk> is a professor at UCL. His lab develops mathematical and experimental tools to understand how learning emerges in brains and artificial neural networks. (Confirmed)

Gautam Reddy <greddy@princeton.edu> is an assistant professor at Princeton, whose lab develops physics-inspired theory and tools for understanding learning and decision-making in biological and artificial systems. (Confirmed)

Shikai Qiu <sq2129@nyu.edu> is a PhD student at New York University whose research focuses on the science of scaling neural networks and optimization.

Kyle Lo <kyleclo@cs.washington.edu> is a member of technical staff at Microsoft AI. Previously, he co-led the OLMo project on open language modeling. His current research focuses on large-scale pretraining of language models, with an emphasis on data curation, data mixing, benchmark validity, efficient experimentation, and generalization across model scales. (Confirmed)

Martin Ziqiao Ma <marstin@umich.edu> is a Member of Technical Staff at Thinking Machines Lab. His research focuses on continual multimodal learning with minimal and natural supervision. (Confirmed)

3 TENTATIVE SCHEDULE AND LOGISTICS

Tentative important dates for submissions, reviews, and final materials: We invite short (5-page) or long (9-page) paper submissions. Submissions will be managed via OpenReview.

- Paper submission deadline: February 1, 2027
- Reviewer bidding and assignment period: February 2 – February 4, 2027
- Reviewer deadline: February 18, 2027

- Acceptance/rejection notification: February 24, 2027
- Accepted papers public on OpenReview and imported to iclr.cc: February 26, 2027
- Camera-ready deadline: March 19, 2027
- Workshop schedule (opening/closing times) posted on iclr.cc: March 26, 2027
- Final workshop program finalized in organizer portal: April 14, 2027
- Workshop date: April 29 or 30, 2027 (TBD by ICLR)

Concurrent submissions. There are no concurrent workshop proposals submitted to other venues.

Tentative schedule. We propose a full-day workshop consisting of invited talks, contributed oral presentations selected from accepted submissions, two poster sessions, and a combined panel and interactive discussion designed to foster community engagement.

Time	Event		Time	Event
9:00–9:10	Opening Remarks		13:30–14:10	Invited Talk 4
9:10–9:50	Invited Talk 1		14:10–14:50	Contributed Orals
9:50–10:30	Invited Talk 2		14:50–15:30	Poster Session 2
10:30–11:15	Poster Session 1		15:30–16:00	Panel Discussion
11:15–11:55	Invited Talk 3		16:00–17:00	Interactive “Debugging Training” Session
11:55–12:45	Contributed Orals		17:00–17:10	Closing Remarks
12:45–13:30	Lunch			

The closing hour combines a moderated panel discussion on open challenges in the field with an interactive “debugging training runs” segment, where participants bring real training bugs—instabilities, divergences, scaling pathologies—for the panelists and audience to diagnose together. We will collect training bug examples through a short survey shortly before the workshop, then split into small-group discussions led by organizers and panelists. In the final 20 minutes, each group will share key takeaways and highlights from their discussion. Following the workshop, we will host an informal social for organizers, speakers, and participants.

3.1 PARTICIPATION AND ACCESSIBILITY PLAN

At ICLR 2027, we anticipate receiving approximately 100 paper submissions, with an expected acceptance rate of around 35–45%. Of the accepted papers, approximately 20–30% will be invited for contributed oral presentations, with the remainder presented as posters. As this is a first-time workshop, prior submission, acceptance, and attendance statistics are not available.

Attendees. We expect 100–200 in-person attendees. Our goal is to attract a diverse range of participants, including: 1) graduate students and early-career researchers, through accessible workshop materials, mentorship-oriented programming, and opportunities for contributed presentations; 2) researchers from underrepresented and interdisciplinary communities, through broad outreach, inclusive call-for-papers language, and targeted dissemination through relevant Slack and Discord communities, EleutherAI channels, social media platforms such as X/Twitter and Bluesky; and 3) senior researchers and practitioners, through invited talks, panel discussions, and opportunities for collaboration.

Virtual Access and Outreach. To broaden participation, we will maintain an online presence through the workshop website, social channels, academic mailing lists, and relevant research communities. Where permitted, accepted papers, posters, and presentation materials will be posted online, along with summaries and key discussions, to support participants unable to attend physically and engage the broader community.

Diversity of Organizers and Speakers. Our organizing team reflects this commitment, spanning diverse career stages, disciplines, institution types, geographies, and lived experiences, including nontraditional academic pathways and underrepresented gender identities. Our proposed speakers and panelists similarly represent a range of seniority levels, research communities, and institutional contexts. As we finalize invitations, we will continue prioritizing diversity across lived experiences and career.

Special Requirements & Technical Needs. No special needs beyond the standard NeurIPS workshop setup. We request poster space for two poster sessions and standard AV support, including projector/display, microphones, reliable internet, live streaming, and recording where supported by NeurIPS.

4 ORGANIZERS AND PROGRAM COMMITTEE

Below are the organizers bio and a list of confirmed program committee. To the best of our knowledge, none of the organizers are listed on any other NeurIPS 2026 workshop proposals. Organizers will not assess submissions from their own institutions or from individuals with whom they hold personal conflicts of interest, and any such submissions will be handled solely by non-conflicted organizers and reviewers.

Isabelle Lee <lee.isabelle.g@gmail.com> is a PhD student at the University of Southern California. Her work focuses on interpretability, training, and reasoning. Specifically, her research aims to (1) predict training and (2) predict failures. To this aim, she uses dynamical systems and physics-inspired approaches to analyze small-scale toy learning problems, as well as analyzing larger-scale training from developmental perspectives. She is a recipient of the Viterbi School of Engineering Graduate Fellowship and Coefficient Giving’s Technical AI Safety Research Grant. Previously, she was a researcher at Carnegie Mellon University and Amazon Alexa, and her background is in Plasma Physics and Complex Systems.

Emmy Liu <emmy@cmu.edu> is a PhD Candidate at Carnegie Mellon University in the Language Technologies Institute. Her current work is on understanding pretraining from a cognitive perspective. She is a recipient of a Natural Sciences and Engineering Research Council of Canada Doctoral Scholarship, the SoftBank-ARM fellowship, an NEC fellowship as well as a CMU Presidential Fellowship. Her work has received the Best Resource Paper award at ACL 2023. She has co-organized the workshop on Figurative Language Processing at NAACL 2024, the overall Student Research Workshop at NAACL 2025, as well as the Computational Developmental Linguistics workshop at ACL 2026.

Kaiser Sun <hsun74@jhu.edu> is a Ph.D. student in Computer Science at the Data Science and AI Institute and the Center for Language and Speech Processing at Johns Hopkins University. Their research focuses on the training dynamics, evaluation, and interpretability of Large Language Models. Generally, Kaiser is interested in understanding how language models function and exploring how we can change them. Their work has been recognized with an honorable mention at CoNLL2023, an outstanding paper award at MLRC2022, and coverage by a few media outlets such as MIT Technology Review and WIRED. Previously, they obtained their B.S. and M.S. from the University of Washington and were a researcher at Meta AI, Together AI, and AWS AI Labs.

Daniel Khashabi <danielk@jhu.edu>

Stella Biderman <stella@eleuther.ai> is the Executive Director of EleutherAI, a non-profit research institute and open science community that does research on large language models, works to make research on large-scale AI technologies more widely accessible around the world, and works to grow and promote the open source AI community. She pioneered the development of open and transparent language models that make the type of research advocated for in this workshop possible, including the first state-of-the-open LLMs to be released alongside their training data (Black et al., 2021; 2022; Gao et al., 2020) and the Pythia model suite (Biderman et al., 2023b; van der Wal et al., 2025) which introduced the idea that releasing partially trained model checkpoints was scientifically valuable and set the standard for maximally open releases adopted by projects such as LLM360, OLMo and Marin. Her lab has done influential research on learning dynamics and interpretability, including introducing Sparse Autoencoders as a tool for interpretability (Cunningham et al., 2023), introducing the problem of forecasting which specific sequences are memorized by a large language model during training (Biderman et al., 2023a), performing the first study of the stability of circuit analysis over the course of training (Tigges et al., 2024), and demonstrating that it is possible to use data filtering to eliminate undesirable capabilities in large language models (O’Brien et al., 2025). Her recent position paper at ICML outlines the need for the type of work solicited in this workshop (Biderman et al., 2026).

Stella co-leads the EvalEval Coalition, a multi-stakeholder initiative bringing methodological rigor and accountability to AI evaluation, is on the advisory board to the Machine Learning Reproducibility Challenge, is organizing an Open-Weight Pre-training Safety Accelerator as part of the Seoul Alignment Workshop (co-located with ICML), and is an AC for NeurIPS. Previously, she was an organizer of the AI Village @ DEF CON (2018-2021), working group lead for the BigScience

Research Workshop (2022), and organized Mozilla and EleutherAI’s Dataset Convening. Stella received her M.S. in Computer Science from Georgia Tech, her S.B. (honors) in Mathematics from the University of Chicago, and her A.B. in Philosophy from the University of Chicago. Her work regularly appears at NeurIPS, ICML, ICLR, and ACL.

David Alvarez-Melis <dam@seas.harvard.edu> is an Assistant Professor of Computer Science at Harvard University, where he leads the Data-Centric Machine Learning (DCML) Lab, with appointments as Associate Faculty at the Kempner Institute and Research Scientist at Microsoft Research. His group develops principled methods for characterizing, curating, and optimizing training data, work that bears directly on the workshop’s prescriptive axis—in particular, treating data selection, ordering, and composition as levers for shaping training trajectories and downstream behavior. He has organized NeurIPS-scale workshops at the intersection of machine learning and applied mathematics, most directly comparable being the Optimal Transport and Machine Learning Workshop at NeurIPS 2023, along with the Optimal Transport Workshop at the Institut d’Études Scientifiques de Cargèse (2024) and the LatinX in AI Workshop at ICML (2023, 2024). David received his PhD in Computer Science from MIT, his MS in Mathematics from NYU’s Courant Institute, and his BSc in Mathematics from ITAM in Mexico City. His work appears regularly at NeurIPS, ICML, ICLR, AISTATS, and UAI.

4.1 PROGRAM COMMITTEE

We are continuing to recruit program committee members with expertise across the workshop themes and will aim for diversity across institutions, geographies, career stages, and research areas. We anticipate approximately 100 submissions. Each paper will receive 3 reviews, with no reviewer assigned more than 3 papers, requiring a program committee of approximately 100 members. **To manage conflicts of interest**, submissions will be assigned so that no reviewer or organizer evaluates a paper from their own institution or from anyone with whom they have a personal conflict (e.g., advisor–advisee, close collaborators, or family), and organizers with a conflict on a given submission will be recused from its decision. Our current program committee (22 confirmed) includes:

Sarah Liaw (Harvard), Millicent Li (Northeastern), Leonardo Blas Urrutia (USC), Terry Zhou (Harvard), Indranil Halder (Harvard), Jennifer Lu (Brown/MIT), Ruochen Zhang (Brown/Microsoft), Rachit Bansal (Harvard), Logan Riggs-Smith (AISST), Alice Rigg (NDIF), Jeffrey Andrade (Harvard), Jay Huang (JHU), Bernal Jimenez Gutierrez (JHU), Niyati Bafna (JHU), Bingbin Liu (Harvard), Marc Marone (JHU), Elisabeth Fittschen (JHU), Leo Du (JHU), Glenn Matlin (Georgia Tech), Xiaochuan Li (CMU), Atharva Naik (CMU), Dabashish Chakraborty (JHU)

We also plan to host an after-social for organizers, program committee, participants and speakers, organized by our social chair, Glenn Matlin (Georgia Tech).

4.2 ADVISORY COMMITTEE

Confirmed: Zhuoran Lu (Hong Kong Univeristy), Jen-Tse Huang (JHU), Jie Gao (Monash University)

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